



M002 Week 8 — Final Cube Maze

Topic: Final Cube Maze · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

Last week. Build a 3D cube maze with a door and a lever. Write code that solves it on its own.

1 · Predict 🧠

```
keep going until at goal:
  if wall ahead OR closed door ahead:
    turn right
  else if lever ahead:
    interact
  else:
    move forward
```

Name one maze where this gets stuck.

2 · Trace

```
keep going until at goal:
  if lever ahead:
    interact
  else if wall ahead:
    turn right
  else:
    move forward
```

Agent sees a lever, then open path, then a wall. What does it do each step?

3 · Fill the Blanks

This loop should solve the maze. Fill in the two blanks.

```
keep going until at goal:
    if wall ahead:
        agent.turn(_____)
    else:
        agent.move(FORWARD, ____)
```

Word bank: LEFT_TURN · RIGHT_TURN · 1

4 · One Bug 🐛

This must use **interact** **only** when a lever is ahead. Right now it interacts every step.

```
keep going until at goal:
  interact
  if wall ahead:
    turn right
  else:
    move forward
```

Write the fixed code.

Why was it wrong?

5 · Build It 📷 + Video Plan 🎥

Build your final maze:

- **20 × 20 × 20** cube
- more than one floor
- 1 redstone door
- 1 lever the agent flips
- start and finish
- code with `while` + `if/else` + `interact` (use `agent.move(UP, 1)` for floors)

Send a photo or video of the agent at the goal. Write 3-4 sentences.

My maze has ... floors and ... levers.

The agent's strategy is ...

The hardest part was ...

I am proud of ...

Phone video plan. Use these words: **cube, floor, lever, door, strategy.**

1. Show the cube outside, then inside.
2. Point to the start, goal, door, and lever.
3. Run your code to the goal.
4. Read your main loop out loud and say what it does.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.